

Nguyen Minh Duc

ducto489.github.io | dustnn00@gmail.com | linkedin.com/in/dustnn

EDUCATION

VNUHCM-University of Science

Bachelors of Science – Honor Program in Physics

Ho Chi Minh City, VN

Sep. 2024 – June 2028

EXPERIENCE

Anyglow (Startup)

Oct. 2025 – Mar. 2026

Machine Vision AI Developer

- Reduced end-to-end pipeline latency from 3.4s to 1.6s ($2.1\times$ speedup) by rewriting data preprocessing and postprocessing with Numba CUDA, moving CPU-bound ops onto the GPU.
- Cut GPU inference time from 1.74s to 1.27s (27% reduction) via post-training quantization (PTQ) from FP16 to INT8 only with 2% drop in mAP50-95 metric using TensorRT and NVIDIA Model Optimizer, with calibration dataset curated from defect distribution analysis.
- Fine-tuned YOLO and EfficientNet on proprietary fabric defect data.

RESEARCH

Flexible Genetic Algorithm for Quantum Support Vector Machines

Nov 24, 2025

First Author

arxiv.org/pdf/2511.19160

- Proposed GA-QSVM: hybrid framework using Genetic Algorithms to automatically optimize quantum feature maps, overcoming limitations of fixed ansatzes (ZZFeatureMap, HEE, IQP).
- Achieved accuracy comparable to classical SVMs and standard QSVMs across four datasets (Digits, Fashion, Wine, Breast Cancer), with transfer learning experiments confirming evolved circuits generalize across tasks.

PROJECTS

Urban Heat Island Classification zindi.africa/zc/YRHymg8

Jan. 2026

Rank 47 / 174 · Top 27% (private leaderboard) - jumped from 100+ public to 47 private ranking.

Solo

- Cross-city geospatial ML: trained on Santiago & Rio de Janeiro (South America), predicted UHI intensity (Low/Medium/High) for Freetown, Sierra Leone (West Africa) using Sentinel-2 + 3D building footprint features.
- Strong public to private rank improvement demonstrates robust cross-domain generalization without overfitting.

Remote Sensing & Signal Processing Project

Sep 2025

Global Nominee for 2025 NASA Space Apps Challenge

Lead & ML Engineer

- Lead the research and adaptation of [state-of-the-art hyperspectral analysis techniques](#) to work with multispectral Sentinel-2 satellite imagery.
- Built a custom unsupervised pipeline using PyTorch, implementing PCA and Gaussian Mixture Model to cluster vegetation growth patterns from time-series multi-channel sensor data (Multispectral Sentinel-2) data.
- Engineered specific spectral features (NDVI_{re1}, Yellowing Index) and a temporal logic algorithm to filter and classify phenological stages with high accuracy.

Vietnamese OCR ducto489.github.io/projects/ocr-dali [Github](#)

May 2025

Solo

- Achieved up to 25% training speedup using NVIDIA DALI, optimizing data processing and augmentations.
- Developed and released a PyTorch Lightning OCR model (ResNet+BiLSTM+Attention) trained on diverse English/Vietnamese data for robust multilingual recognition.
- Engineered and open-sourced 11M+ OCR images datasets with character normalization for better learning.

Train GPT-2 with TPU ducto489.github.io/projects/train-gpt-2-with-tpu/

Aug. 2024

Solo

- Reproduced the GPT-2 124M based on GPT-2 and GPT-3 paper on Kaggle.
- Implemented gradient accumulation, distributed data parallel (GPU and TPU), half-precision, and flash attention.
- Sped up training by 33 times compared to GPU T4 x2 using TPU, BF16, and some other TPU optimization.
- Surpassed GPT-2 result with validation loss 3.2754 over 3.2924 and HellaSwag evaluation 0.2962 over 0.294463.

- Designed an Advantage Actor-Critic (A2C) agent to solve discrete optimization problems with large action spaces (2,315 actions), utilizing stochastic policy gradients analogous to robotic control strategies.
- Using transfer learning from 10 words and progressively expanded to 100 words and the full 2,315 words.

AWARDSTop 2% of users on [Kattis Problem Archive](#)

Feb. 2024

- Solved numerous algorithmic challenges including data structures, dynamic programming and graph theory.

SKILLS**Programming:** Python, C, C++, CUDA**Libraries & Frameworks:** PyTorch, PyTorch Lightning, TensorRT, NVIDIA DALI, Numba CUDA, NVIDIA Model Optimizer, Hugging Face, Scikit-learn, NumPy, Pandas, OpenCV, Qiskit**Techniques:** Post-Training Quantization (INT8), Distributed Data Parallel, Mixed Precision (BF16/FP16), Flash Attention, Transfer Learning, A2C, Genetic Algorithms**Domains:** Computer Vision (YOLO, EfficientNet, OCR), Quantum ML (QSVM, Variational Circuits), Remote Sensing (Sentinel-2, VIIRS, MODIS, NDVI)**Languages:** English, Vietnamese